Floe Foxon *Scientist·Analyst·Programmer*

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EXPERIENCE

Scientist and Data Analyst Statistician	Pinney Associates, Inc.	2024–Present 2020–2024
Research Scholar	Sanford Research	2019
Python Programming Tutor	CodeX, Enactus Nottingham	2018

EDUCATION

MSc Data Science (Statistics) Courses: Programming for Data Science Statistical Computing Bayesian Statistics Machine Learning	University of Leeds	2024–2025
GCert Artificial Intelligence and Machine Learning Courses: AI in Astrophysics Machine Learning AI in Healthcare Deep Learning	University of Texas at Austin	2026–2027
BSc (Hons) Physics with Astronomy Courses: Computing for Physical Science Mathematics for Physics Scientific Computing	University of Nottingham	2017–2020
General Certificate of Education Advanced Level Subjects: Computer Science Mathematics Physics	Bishop Vesey's Grammar School	2015–2017

PROFESSIONAL DEVELOPMENT

Wolfram Summer Research Institute, Physics and Foundational Science Track	Wolfram Research, Inc.	2025
Data-Driven Astronomy Certificate	University of Sydney	2024
Professional Certificate Data Science	IBM	2020

SELECTED WORKS

Foxon, F., Sembower, M.A., Shiffman, S. 2026. Temporal Generalized Linear Mixed Models to Monitor National Health Behaviour Using Subnational Observational Data. *Under Review.*

Foxon, F. 2026. Statistical Tests for Randomness on a Typewritten Key Stream Extracted With Computer Vision and Classified With a Convolutional Neural Network. Proceedings of the 9th International Conference on Historical Cryptology HistoCrypt 2026. *Accepted - In Press.*

Foxon, F. 2025. Mining Pockets of Computational Reducibility with AI: Transformer Models of Graph Rewriting Systems. Wolfram Community, Staff Picks.

Foxon, F. 2025. Solar Cycles: Can They Be Predicted? Research Notes of the American Astronomical Society. 9(2):40.

SELECTED WORKS (CONTINUED)

Foxon, F. 2025. Machine Learning for Text Classification in Classical Cryptography. Proceedings of the 8th International Conference on Historical Cryptology HistoCrypt 2025. 60-64.

Foxon, F. 2024. Artificial Neural Network for Hoax Cryptogram Identification. Proceedings of the 7th International Conference on Historical Cryptology HistoCrypt 2024. 86-90.

Foxon, F. 2023. Can Bayesian Statistics Be Used to Analyze Phenomena in Folk Zoology? Journal of Scientific Exploration. 37(3):570–571.

Foxon, F. 2021. Evaluating Modern System Dynamics Software for Use in Astrophysical Simulations. Astronomy and Computing. 36:100486.

Foxon, F. 2021. Ammonoid Taxonomy with Supervised and Unsupervised Machine Learning Algorithms. PCI Paleontology.

AWARDS

Distinction Award	University of Leeds	2026
Leeds Partnership Awards Nominee (×2), Shortlisted	University of Leeds	2026
Chambliss Student Poster Competition Finalist	American Astronomical Society	2025
First Class Honours Award	University of Nottingham	2020
First Year Scholarship Award for Academic Achievement	University of Nottingham	2018

SKILLS

Programming	Python (scikit-learn, PyTorch, TensorFlow, pandas, Matplotlib), Jupyter Notebook, R, SQL, SAS, MATLAB, Java, Wolfram
Software	Microsoft Office 365/SharePoint, Google Workspace, Linux, Windows, GitHub, Tableau, Mathematica

SERVICES

Chair/Host	Folk Zoology Conference 2024 (Virtual); Folk Zoology Conference 2026 (Virtual)
Ambassador	University of Nottingham (2018–2019); Students for Sensible Drug Policy (2026)
Peer Reviewer	Scientific Reports; PeerJ; American Journal of Health Behavior; Drug and Alcohol Dependence; Harm Reduction Journal; Internal and Emergency Medicine; Contributions to Tobacco & Nicotine Research; Addictive Behaviors; SSM - Qualitative Research in Health; Nicotine & Tobacco Research

RESEARCH INTERESTS

I am passionate about all sciences and mathematics and my personal goal is to solve complicated problems. I look forward to working on innovative scientific and technical projects that meet cost and time budgets. In particular, I am interested in applications of machine learning and modelling techniques to novel data.

SELECTED MEDIA APPEARANCES

Science ◇ New Scientist ◇ Retraction Watch ◇ Popular Mechanics ◇ Photonics Focus